

Real Estate Market Insights

Exploratory Data Analysis of Zameen.com Listings in Pakistan
AtomCamp Data Analytics Bootcamp — DA10

1. Problem Statement

Pakistan's real estate market is large but poorly transparent for retail investors. Zameen.com is the country's leading property portal with hundreds of thousands of listings. This report summarizes an end-to-end exploratory analysis of 18,255 property listings.

Business Question: What drives property prices in Pakistan — and how do prices vary across cities, property types, and sizes?

2. Data Understanding & Preprocessing

Only the 8 most relevant columns were imported: City, Location, Price, Type, Area, Purpose, Bedrooms, Bathrooms.

Attribute	Detail
Raw rows	18,255
Columns loaded	8
Duplicate rows found	3,826 (21%)
Rows after deduplication	14,429
Unique cities (raw)	84
Unique property types	7
Price range	PKR 6,000 – PKR 1.8 Billion

3. Missing Values Treatment

After deduplication, only 56 missing values were found — 8 per column — representing less than 0.1% of the data.

Column	Missing	Treatment	Justification
City	0	None needed	Fully populated
Type, Purpose, Location	8 each	Mode imputation	Most frequent category for string columns
Bedrooms, Bathrooms	8 each	Median imputation	Avoids non-integer values like 2.3 bedrooms
Price, Area	8 each	Cleaned in next step	Require unit parsing before imputation

4. Data Cleaning & Consistency

Five cleaning steps were applied:

- **City standardization:** 84 raw city variants were mapped to 6 canonical cities using FuzzyWuzzy string matching.
- **Price parsing:** Removed PKR symbols and newlines; converted Lakh, Crore, and Arab units to a plain numeric PKR float.
- **Area conversion:** Marla ($\times 272.25$), Kanal ($\times 5,445$), and Sq.Yd ($\times 9$) all converted to uniform square feet.

- **Purpose cleaning:** FuzzyWuzzy was applied to standardize Purpose variants. After cleaning, all 14,429 records resolved to 'For Sale' — the 'For Rent' category produced no matches above the 60% threshold, indicating the dataset contains only sale listings.
- **Outlier capping:** $IQR \times 3.0$ upper fence = PKR 139.6M. 567 extreme outliers (3.9%) were capped rather than dropped to preserve record count.

City	Listing Count
Islamabad	3,660
Rawalpindi	2,883
Karachi	2,530
Lahore	2,254
Faisalabad	1,741
Multan	1,361

5. Feature Engineering

Four new columns were derived to enable richer analysis:

Feature	Formula	Purpose
Price_per_sqft	Price ÷ Area	Fair comparison across different property sizes
Log_Price	$\log(1 + \text{Price})$	Handles right-skew in visualizations
Bedroom_cat	Bins: Studio, 2-Bed, 3-Bed, 4+-Bed	Categorical lens for price segmentation
Area_cat	Bins: Compact / Standard / Large / Estate	Groups properties by scale

6. Univariate & Bivariate Analysis

6.1 Price Distribution

Raw prices are heavily right-skewed — the mean (PKR 25.1M) is 1.83× the median (PKR 13.7M) after outlier capping. After log-transformation, the distribution becomes near-normal, confirming that property prices are log-normally distributed. Percentage-based comparisons are more meaningful than absolute differences.

6.2 Listing Volume

Houses are the most common property type with 10,181 listings (70.6% of all records after cleaning). Flats are second with 2,603 listings, followed by Upper Portion (989), Lower Portion (498), Farm House (87), Room (41), and Penthouse (22). By city, Islamabad has the most listings (3,660) followed by Rawalpindi (2,883). All 14,421 records (after dropping 8 rows with missing Price/Area) resolved to For Sale after Purpose cleaning — the dataset contains no For Rent listings.

6.3 Median Price by City (For Sale)

City	Median Price (PKR)	Relative Ranking
Multan	17.5 Million	Highest
Faisalabad	17.0 Million	2nd
Islamabad	15.0 Million	3rd

Rawalpindi	12.0 Million	4th
Karachi	10.0 Million	5th
Lahore	6.3 Million	Lowest

6.4 Price per Sqft by City

When normalized by area, Faisalabad commands the highest median price-per-sqft (PKR 12,121), followed by Karachi (PKR 9,986), Islamabad (PKR 9,917), and Multan (PKR 9,917). Lahore has the lowest median price-per-sqft (PKR 5,969), despite having lower total listing prices — suggesting Lahore listings tend to be larger properties at lower per-sqft rates.

6.5 Price by Bedroom Category

Bedroom Category	Median Price (PKR)
Studio / 1-Bed	120,000
2-Bed	152,500
3-Bed	280,000
4+-Bed	24.0 Million

The bedroom price distribution reveals a sharp two-tier market: Studio, 2-Bed, and 3-Bed properties have very low median prices (PKR 120K–280K), while 4+-Bed properties jump dramatically to PKR 24M. This suggests that smaller bedroom-count listings in this dataset are predominantly plots or low-cost units, while 4+-Bed properties represent full residential houses commanding premium prices.

6.6 Correlation Heatmap

Feature Pair	Correlation (r)	Interpretation
Area ↔ Price	0.13	Weak positive — size alone is not the main driver
Bedrooms ↔ Price	0.42	Moderate positive
Bathrooms ↔ Price	0.42	Moderate positive
Bedrooms ↔ Bathrooms	0.80	Very strong — multicollinear, use only one in models
Price_per_sqft ↔ Price	0.21	Weak positive — location adds some premium beyond size

7. Key Insights & Recommendations

Findings

- City is a significant price driver, but not in the expected direction — Multan and Faisalabad have the highest median total prices (PKR 17.5M and 17.0M), while Lahore has the lowest (PKR 6.3M). However, Faisalabad leads on price-per-sqft (PKR 12,121), suggesting Lahore has larger but cheaper-per-sqft properties.
- Bedrooms is the strongest numeric correlate with price (r=0.42). Area has a surprisingly weak correlation (r=0.13), likely due to high variance in area units and property mix.
- Property prices follow a log-normal distribution — always use medians rather than means for analysis. The mean (PKR 25.1M) is 1.83x the median (PKR 13.7M) after outlier capping.
- Bedrooms and Bathrooms are highly multicollinear (r=0.80). Use only one as a predictor in any model.
- A sharp two-tier pricing gap exists: Studio through 3-Bed properties have median prices of PKR 120K–280K, while 4+-Bed properties jump to PKR 24M — indicating two distinct market segments.

Investor Recommendations

Strategy	Rationale
Target Multan & Faisalabad for high total price listings	These cities show the highest median total prices (PKR 17.5M and 17.0M) in this dataset
Use price-per-sqft for fair city comparison	Faisalabad leads at PKR 12,121/sqft while Lahore is lowest at PKR 5,969/sqft — opposite of total price
Focus on 4+-Bed properties for premium segment	Median price jumps from PKR 280K (3-Bed) to PKR 24M (4+-Bed) — a clear luxury threshold
Do not rely on area alone to predict price	Area-Price correlation is only 0.13 — bedroom count (r=0.42) is a better price predictor in this data
Dataset is 100% For Sale listings	All records resolved to For Sale after cleaning — rental analysis requires a separate data collection

8. Conclusion & Next Steps

This analysis of 18,255 Zameen.com listings successfully transformed raw, messy listing data into a clean analytical dataset through a rigorous 5-step cleaning pipeline. Contrary to conventional expectation, area size has a weak correlation with price (r=0.13) in this dataset, while bedroom count (r=0.42) is a stronger predictor. Multan and Faisalabad show the highest median total prices, while Faisalabad leads on price-per-sqft. A sharp two-tier gap between small (Studio–3 Bed) and large (4+-Bed) properties defines the market segmentation.

Suggested Next Steps

- Train a regression model (Ridge, XGBoost) to predict price from City, Area, Type, and Bedrooms.
- Perform sub-city neighborhood analysis using the Location column (e.g., DHA vs Bahria Town).
- Collect For Rent listings from Zameen.com to enable gross rental yield calculation (Annual Rent / Sale Price) by city and property type.
- Apply NLP on the Description column to extract features (gated community, near school) that command premiums.
- Add time-series analysis if listing dates are available to track price appreciation over time.